

*MA.7.GR.2.1***Benchmark**

MA.7.GR.2.1 **Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.**

Benchmark Clarifications:

Clarification 1: Instruction focuses on representing a right circular cylinder with its net and on the connection between surface area of a figure and its net.

Clarification 2: Within this benchmark, the expectation is to find the surface area when given a net or when given a three-dimensional figure.

Clarification 3: Within this benchmark, the expectation is not to memorize the surface area formula for a right circular cylinder.

Clarification 4: Solutions may be represented in terms of pi (π) or approximately.

Connecting Benchmarks/Horizontal Alignment

- MA.7.GR.1.3
- MA.7.GR.2.3

Terms from the K-12 Glossary

- Cylinder (Circular)
- Net
- Pi (π)
- Surface Area

Vertical Alignment**Previous Benchmarks**

- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.GR.1.2
- This is the first introduction to cylinders outside of recognizing the shape in early elementary.

Next Benchmarks

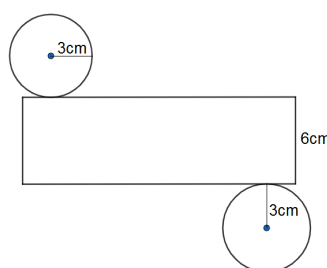
- MA.912.GR.4.6

Purpose and Instructional Strategies

In previous courses, students found the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles, which developed into finding the surface area of right rectangular prisms and right rectangular pyramids using a figure's net. In grade 7 accelerated, students find the area of a circle as well as the surface area of a right circular cylinder using the figure's net. In future courses, students will solve mathematical and real-world problems involving the surface area of cylinders, pyramids, prisms, cones and spheres.

- Instruction includes problems that make connections to the understanding of a formula for the surface area of a right circular cylinder and its net (soup can and finding the area of the paper around the can). Show how different parts of the formula correspond to different parts of the net (*MTR.7.1*).

- Instruction allows students the opportunity to develop the formula for the surface area of a right circular cylinder.
 - For example, provide students cans of various sizes, paper, scissors, and tape (*MTR.2.1*). Ask students to cover the can completely without overlapping the paper and only using as little tape as possible. Ask students to explain their method if they feel it was successful in covering the can completely without overlapping any paper, until someone presents the idea of cutting two circles, taping them to the base and then one rectangle to cover the curved surface (*MTR.4.1, MTR.5.1*).
- Instruction allows students to use various nets for cylinders which they can cut out and form the three-dimensional figure or use virtual simulations which show the “unrolling” of the cylinder into its net (*MTR.2.1*).



- Instruction focuses on recognizing the height of a cylinder as the perpendicular distance between the parallel (circular) bases.

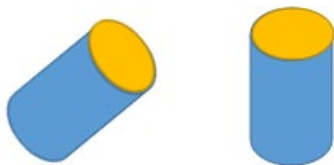
Common Misconceptions or Errors

- Students often confuse the vocabulary base, length, height and “*B*” (base area), when moving between two- and three-dimensional figures.
- Students may incorrectly invert the height and base of a cylinder when the cylinder is presented horizontally. To address this misconception, remind students that while a cylinder may lay on its side, the bases are the parallel circles.
- Students may incorrectly apply the formulas for area, surface area and volume. To address this misconception, use these concepts in context, or a manner in which they understand the meaning behind the terms to foster their conceptual development (*MTR.7.1*).

Strategies to Support Tiered Instruction

- Instruction includes the use of geometric software to allow students to explore the difference between base, length, height and “*B*” (base area).
- Instruction includes co-creating a graphic organizer to define the dimensions of rectangles, circles and right circular cylinders.
- Teacher provides students with an example of a three-dimensional figure in its original position then provides multiple orientations to discuss how the location of the figure’s base changes, but the dimensions of the figure do not change.

- For example, two right circular cylinders are shown below with the same dimensions but in different orientations. A base is highlighted in each.



- Teacher instructs students to draw a visual of a three-dimensional figure and its dimensions in the context of a real-world problem.
- Instruction includes finding the exact area of a given circle in terms of π before replacing the value of π with an approximation to determine the estimated area.
- Instruction includes color-coding and labeling the dimensions of rectangles, circles and right circular cylinder.
- Teacher provides instruction using manipulatives or geometric software for students to understand the difference between the formulas for area, surface area and volume.
- Teacher encourages students to continue to use the parts of the net to calculate the surface area, rather than focusing on a formula.
- Teacher uses formulas for area, surface area and volume in context, or a manner in which students can understand the meaning behind the terms.
- Provide multiple orientations of cylinders, focusing on the height as the perpendicular distance between the parallel bases, continuing to break them down to their nets.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

An ocean resort decided to build a large room in the shape of a cylinder to host events. The room is 34 feet in diameter with a height of 9 feet. They are going to paint the floor, wall and ceiling blue to make attendees feel like they are floating in the sky. Determine the surface area that needs to be painted.

Instructional Task 2 (MTR.7.1)

Reviews from several events in a blue room at a resort have come in and attendees are reporting feeling trapped in the enclosed room. The room is 34 feet in diameter with a height of 9 feet. The resort has decided to replace the solid wall with windows for a 360° view of the ocean and surrounding area.

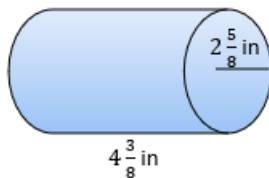
Part A. How many square feet of windows will need to be ordered to do so?

Part B. If the resort decided to make a glass ceiling instead of replacing the curved wall, how much glass would be needed?

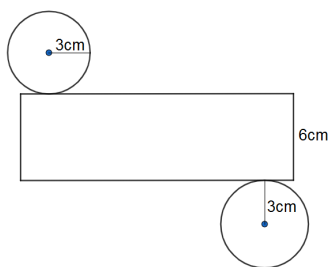
Instructional Items*Instructional Item 1*

Determine the surface area of the cylinder below. Round to the nearest tenth.

Note: Figure is not drawn to scale.

*Instructional Item 2*

Determine the surface area of the cylinder below. Write your answer as the exact surface area.

*Instructional Item 3*

Lyndon is making a nylon case for his new snare drum which measures 14 inches in diameter and is 6 inches deep. If the case fits snugly around the drum, how much nylon will Lyndon need?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*